Naga Sai Dattu

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EDUCATION

Vellore Institute of Technology, Vellore *2023 – 2027*
B.Tech – Computer Science and Engineering (Core) | CGPA: 9.37

FIITJEE Junior College, Vizag *2021 – 2023*
Intermediate (Class XI & XII), MPC Stream | Percentage: 95.7%

PUBLICATIONS / RESEARCH

First Author – A Systematic Survey of IoT-Based Fall Detection and Health Monitoring for Elderly Care
Accepted at ICCET 2026 – International Conference on Contemporary Engineering and Technology, Chennai (pending presentation)

TECHNICAL SKILLS

Languages: C++, Python, Java, JavaScript, SQL, HTML, CSS

Problem Solving: Arrays, Linked Lists, Stacks, Queues, Trees, Binary Search, Sliding Window · Solved 160+ problems on LeetCode

Machine Learning & Data Science: NumPy, Pandas, Matplotlib, Scikit-learn, XGBoost, Streamlit, Spacy, sentence-transformers (NLP)

Web Technologies & Frameworks: React, Node.js, Express.js, EJS, MaterialUI, Bootstrap

Databases & Tools: MongoDB Atlas, MySQL, REST APIs, Git, GitHub, Docker

PROJECTS

CrisisLens — Real-Time Crisis Intelligence System *March 2026*

- Engineered a five-stage NLP pipeline for real-time disaster intelligence: a TF-IDF + LinearSVC classifier filters ~90% noise in under 1 ms/tweet, MiniLM-L6-v2 sentence embeddings with DBSCAN (eps=0.40 cosine distance) deduplicates reports into incident clusters, dslim/bert-base-NER extracts location entities via async ThreadPoolExecutor inference, a 171-location Chennai gazetteer resolves coordinates through fuzzy matching, and BART (facebook/bart-large-cnn) via HuggingFace Inference API generates situation summaries — all orchestrated through a FastAPI backend with in-memory cluster state and REST polling endpoints.
- Built a React "War Room" dashboard featuring a live incident cluster feed with urgency badges (CRITICAL/HIGH/MODERATE), an interactive Leaflet map with geocoded markers, and auto-refreshing BART situation reports. Deployed the backend on an Azure VM and the frontend on Azure Static Web Apps.

GitHub: github.com/Nagasai-Datta/CrisisLens

AI-Powered Stock Trading Platform *October 2025*

- Architected and deployed a dual-application MERN-stack platform (marketing frontend on Vercel + Node.js/Express.js backend on Render) with secure RESTful APIs via Passport.js authentication and MongoDB for all user, order, and portfolio data.
- Engineered an AI-powered portfolio assistant that ingests live holdings and real-time market data, sending contextual prompts to the Mistral AI API to deliver data-driven buy/sell suggestions — enabling informed trading decisions within the dashboard.

GitHub: github.com/Nagasai-Datta/AI_Powered_Stock_Trading_App

Healthcare Premium Predictor *December 2025*

- Improved premium prediction accuracy from 78% R^2 (linear regression baseline) to 98.77% R^2 by implementing XGBoost models trained on age-segmented cohorts (≤25 and >25), capturing distinct risk profiles that a single model could not represent.
- Built a modular prediction pipeline using Joblib to load serialized models and scalers for real-time preprocessing, deployed end-to-end on Streamlit Cloud with a functional UI for seamless data input and results display.

GitHub: github.com/Nagasai-Datta/health_care_premium_predictor

CERTIFICATIONS

GEN AI Using IBM Watsonx — IBM Career Education Program
Git Essential Training — LinkedIn Learning